

Name: _____

Roll No: _____

Division of Computer Engineering
B-Tech Computer Science & Engineering
Fourth Semester Second Internal Examinations, July-2024
19-202-0403 COMPUTER ARCHITECTURE AND ORGANISATION (A-Batch)

Time: 2hrs

Max mark: 50

*Use only Pen to draw tables and figures
Leave a margin of 1" at left side of the answer sheet
Suitably assume any missing data*

PART-A

Attempt ALL questions

1. (a) What is the benefit of using biased representation for the exponent portion of a floating-point number? Provide the representation of the following numbers in IEEE 32-bit floating point representation .
(i) -1.5 (ii) -8 (iii) 384 (iv) 1/16 . (CO4) (4)
- (b) Compare memory-mapped I/O and I/O-mapped I/O (CO8) (4)
- (c) What is cycle stealing ? How it can be utilised in DMA implementation ? Why DMA access to main memory is given higher priority than CPU access to main memory ? (CO8) (4)
- (d) What is the use of TLB? How does it improve the performance of the CPU ? (CO7) (4)
- (e) A pipelined processor has a clock rate of 2.5 GHz and executes a program with 1.5 million instructions. The pipeline has five stages, and instructions are issued at a rate of one per clock cycle. What is the speedup of this processor for this program compared to a non-pipelined processor ? (CO1) (4)

PART-B

Attempt ANY THREE questions

2. (a) Show how the following floating-point subtractions are performed (where significands are truncated to 4 decimal digits). Show the results in normalized form.
(i) $7.744 \times 10^{-3} - 6.666 \times 10^{-3}$
(ii) $8.844 \times 10^{-3} - 2.233 \times 10^{-1}$ (CO5) (5)
- (b) Draw a flow chart showing the addition and subtraction of floating point numbers. (CO4) (5)

P.T.O

3. (a) Draw a flow chart showing Booth's algorithm. Multiply 010111 by 110101 (both numbers are represented in 2's complement) using Booth's algorithm. Show the values in the registers in each step in tabular form. (CO4) (5)
- (b) Show an algorithm for restoring 2's Complement division. Illustrate this algorithm by showing the division of -145 by 13 (CO4) (5)
4. (a) What is the need of a bus arbitrator in a shared I/O bus structure? Explain any one bus arbitration mechanism. (CO8) (5)
- (b) What is daisy chaining? How it can be effectively utilised to handle simultaneous interrupt requests ? (CO8) (5)
5. (a) Draw and explain the architecture of PCI Local Bus. (CO8) (8)
- (b) Why SCSI HDD and SAS drives are widely used on server machines? (CO8) (2)

* All The Best *
Faculty: Vinod Kumar.P.P

Course Outcomes:

1. Acquire knowledge about structure, functions and characteristics of computer systems.
2. Identify the addressing modes used in instructions.
3. Determine the set of control signals generated and their timing sequence, given an instruction.
4. Demonstrate how addition, multiplication and division operations are implemented inside a computer system.
5. Explain each level of memory hierarchy.
6. Show how cache mapping affect the location of the data and the replacement policies.
7. Map a virtual address to physical address.
8. Identify and compare different methods for computer I/O.